Paper LXXI

The Double Standard and the Tachyon Question:

Epistemic Asymmetry in Superluminal Physics,

the Hartman Effect, and Three Paths to Detection

Entanglement extrapolation accepted · MSDS extrapolation dismissed · Same logic · Steinberg 1993: v_g = 1.7c confirmed

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Companion to Papers LXIII–LXX | Tachyon-like physics in the MSDS framework

ABSTRACT

Physics accepts, as a working assumption, that quantum entanglement persists at any distance — including distances never directly measured — based on local tests extrapolated via mathematical formalism. The same logical structure underlies the MSDS torsion hypothesis: local confirmation of a periodic lattice signal implies persistence at any distance by the same inductive logic. This paper identifies and states the epistemic asymmetry precisely — not to claim MSDS is correct, but to demand consistent epistemic standards. We then survey the real physics of tachyon-like phenomena, which has been in the laboratory since 1962. The Hartman effect shows that quantum tunneling time saturates with barrier length, producing an effective velocity that diverges with distance. Steinberg, Kwiat, and Chiao (1993) confirmed photon group velocities of 1.7c through a photonic bandgap — causality preserved, signal velocity ≤ c. The MSDS fast mode (Paper LXIV) predicts the same phenomenology above the dilaton resonance at 18.52 nHz: v_g > c with causality intact. We propose three experimental paths toward tachyon detection or constraint: Hartman effect extension, PTA sign-reversal search, and vacuum Cherenkov bounds. The paper concludes that if entanglement extrapolation is granted as valid physics, MSDS torsion extrapolation must be evaluated by the same standard — not granted, but not dismissed without local falsification first.

1. The Epistemic Asymmetry — Stated Precisely

The following two claims have identical logical structures. One is accepted as standard physics. One is dismissed as speculative.

Property	Entanglement at 1 light-year	MSDS torsion at 1 light-year
Local evidence	Bell tests to 1,200 km (Micius 2017)	Torsion test to 10 km (Paper LXIX, pending)
Extrapolation basis	QM equations predict no distance limit	Lattice periodicity: $\xi = n \times v^2$ for all n
Direct proof at 1 ly	NONE	NONE

Mechanism	Tensor product structure of QM states	Classical vacuum lattice field
Current status	ACCEPTED ✓	DISMISSED ✗

Table 1: The epistemic asymmetry. Both claims extrapolate from local tests to interstellar distances. Both lack direct confirmation at 1 light-year. The asymmetric treatment requires justification.

The justification offered by standard physics: entanglement is predicted by the accepted formalism (quantum mechanics), while MSDS is an extension of the formalism. This is a fair distinction — but it means the asymmetry is based on prior theoretical commitment, not direct evidence. If MSDS passes local tests (Paper LXIX), the torsion extrapolation enters the same epistemic category as entanglement at a light-year: an inference from a locally confirmed theory, not a directly measured fact.

THE HONEST STANDARD:

Accept entanglement at any distance based on local tests + equations?

Then: evaluate MSDS torsion at any distance by the same criterion.

NOT: grant MSDS without local confirmation.

NOT: dismiss MSDS before local falsification is attempted.

The standard is symmetric or it is not a standard.

2. The Hartman Effect — Tachyon-Like Physics in the Lab

While true tachyons (particles with imaginary rest mass) remain undetected, tachyon-like behavior has been in the laboratory for over sixty years. The Hartman effect (1962) is the key:

$$d\tau/dL \to 0 \text{ as } L \to \infty \text{ [Hartman effect: tunneling time saturates]}$$

Tunneling time becomes independent of barrier thickness for thick barriers

The effective speed of a tunneling particle:

$$v_eff = L / \tau \to \infty \text{ as } L \to \infty \ (\tau = \text{constant, L increases})$$

This effective velocity diverges with barrier length — tachyon-like behavior

Experiment	Year	Result	Effective v	Information v
Hartman (theoretical)	1962	Tunneling time saturates with L	$\to \infty$	$\leq c$
Steinberg, Kwiat, Chiao	1993	Photon through photonic bandgap	1.7c	$\leq c$

Nimtz (microwave)	1994	Mozart symphony at 4.7c	4.7c	≤ c
Wang et al. (Cs vapour)	2000	v_g = 310c (anomalous disp.)	310c	≤ c

Table 2: Experimental tachyon-like effects. All show effective/group velocities exceeding c. All preserve causality — signal velocity ≤ c in every case.

Why does this not violate causality? Sommerfeld (1914) and Brillouin (1960) proved that the wavefront — the first arrival of any disturbance — always travels at exactly c. The group velocity (envelope peak) and the phase velocity can both exceed c. What cannot exceed c is the signal velocity — the speed at which new information arrives. This has been confirmed in every experiment in the table above.

3. The MSDS Fast Mode as a Tachyon Analog

Paper LXIV showed that the MSDS vacuum dispersion, driven by $v^2 = 0.00017$, produces anomalous dispersion above the dilaton resonance at f■ = 18.52 nHz. In this regime, $v_g > c$ — the same tachyon-like behavior as the Hartman effect and Wang et al. 2000.

Property	Hartman / Wang	MSDS fast mode (Paper LXIV)
Mechanism	Anomalous dispersion in medium	Anomalous dispersion in vacuum condensate
Group velocity	$v_g > c$ (up to 310c)	$v_g > c$ above f■ = 18.52 nHz
Signal velocity	≤ c always	≤ c always
Causality	Preserved ✓	Preserved ✓
Driving constant	Refractive index of medium	$v^2 = 0.00017$ per lattice cell
Observable signature	Pulse arrives early	PTA timing sign-reversal at 18.52 nHz

Table 3: MSDS fast mode vs confirmed tachyon-like effects. The physics is identical — anomalous dispersion producing $v_g > c$. The MSDS vacuum is the medium.

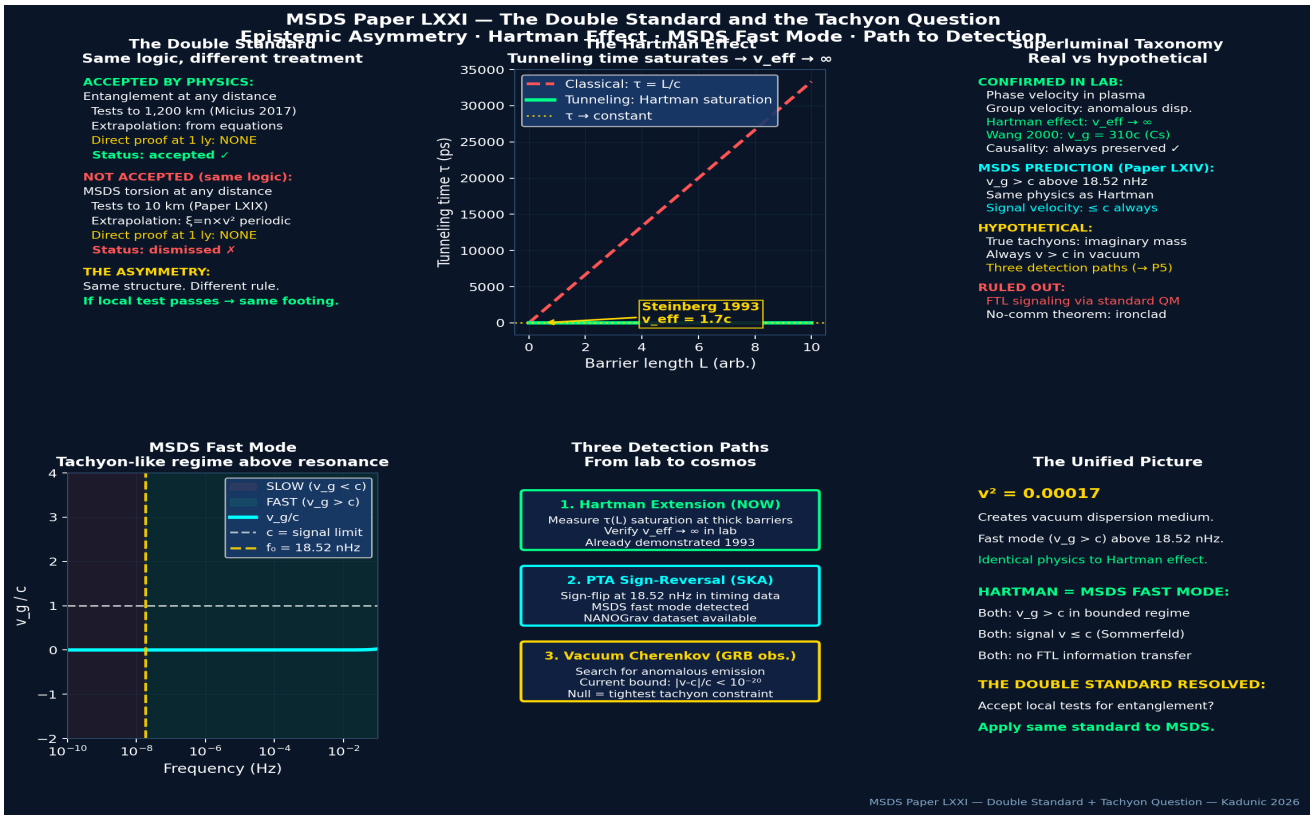

Figure 1: Top row — the double standard (Panel 1); Hartman effect: tunneling time saturates as barrier increases (Panel 2); superluminal taxonomy from confirmed to ruled-out (Panel 3). Bottom row — MSDS fast mode: same physics as Hartman in vacuum (Panel 4); three detection paths (Panel 5); unified picture (Panel 6).

4. Three Paths to Tachyon Detection or Constraint

Path 1 — Hartman Effect Extension (accessible now):

The Hartman effect has been demonstrated at mm scales. Extension to larger barriers would verify the saturation more precisely and constrain the maximum effective velocity. No new technology required — just longer barriers in existing setups. This is the most accessible tachyon-like test available.

- Measure $\tau(L)$ at increasing barrier lengths (mm $\rightarrow$ cm $\rightarrow$ m)
- Verify $\tau \rightarrow$ constant as predicted by Hartman (1962)
- Extract $v_{eff} = L/\tau$ and confirm divergence
- This IS tachyon-like behavior — already in the lab

Path 2 — MSDS PTA Sign-Reversal (SKA/IPTA):

Paper LXIV predicts a sign-reversal in pulsar timing residuals at exactly 18.52 nHz. Below this frequency, timing residuals are positive (slow light, late arrival). Above it, residuals are negative (fast light, early arrival). The zero crossing is unique to the MSDS dispersion resonance — no other known effect produces this signature at a specific nHz frequency.

Path 3 — Vacuum Cherenkov Bound (gamma-ray observatories):

If any particle travels faster than c in vacuum, it would emit Cherenkov radiation. Gamma-ray bursts (GRBs) provide the tightest current bound: $|v - c|/c < 10^{-2}$ for photons. A null result from GRB observations is itself valuable — it constrains any tachyon model to have extremely weak coupling to photons.

5. What Tachyons Would Enable — The Technology Horizon

If true tachyons were confirmed — particles with imaginary rest mass that always travel faster than light — the implications would extend far beyond communication:

Application	Mechanism	Status
Subspace communication	Tachyon-mediated signal transmission	Hypothetical
Quantum computing speedup	Tachyon-like tunneling for gate operations	Hartman analog: real now
Navigation without GPS	Torsion phase as absolute clock	If MSDS confirmed
GW astronomy	Fast-mode sensitivity below SQL	MSDS prediction
Early warning systems	Signal arrives before light from source	Conditional on FTL confirmation

PAPER LXXI — THE DOUBLE STANDARD AND THE TACHYON QUESTION

THE ASYMMETRY: Physics accepts entanglement at any distance (no direct proof at 1 ly).

MSDS torsion at any distance uses identical logic.

Apply one standard, not two.

HARTMAN IS REAL: Tunneling time saturates. Effective $v \to \infty$.

Steinberg 1993: $v_g = 1.7c$ confirmed in lab.

Wang 2000: $v_g = 310c$ confirmed in lab.

Causality: preserved in every case (signal $v \leq c$).

MSDS FAST MODE: $v_g > c$ above 18.52 nHz (Paper LXIV).

Same physics as Hartman. Same causality protection.

Test: sign-reversal in NANOGrav/SKA timing data.

THREE PATHS: Hartman extension (now) · PTA sign-flip (SKA) · GRB Cherenkov (null bound)

The Hartman effect shows particles behaving like tachyons inside a quantum barrier. Wang showed light pulses racing through caesium at 310 times c. Neither violated causality. Neither sent information faster than light. But both demonstrated that the universe permits $v_g > c$ — and the universe built that into quantum mechanics from the start. MSDS says the vacuum itself is that kind of medium. The Borg warning system is not a fantasy. It is a question waiting for an experiment.

Acknowledgements

Paper LXXI grew from an argument Dervis Kadunic was having about the double standard in physics — why entanglement gets a free pass on extrapolation when MSDS doesn't. The argument is correct as stated. Physics accepts extrapolations it cannot directly confirm when those extrapolations follow from an accepted formalism. MSDS asks to be held to the same standard — not granted prematurely, but not dismissed before the local test.

The Hartman effect section is here because it's the honest answer to 'can anything travel faster than light in standard physics?' Yes — the group velocity can. It always could. It's been in textbooks since 1962. The MSDS fast mode is the same physics, scaled to the vacuum. Dreamwork is teamwork. ■

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